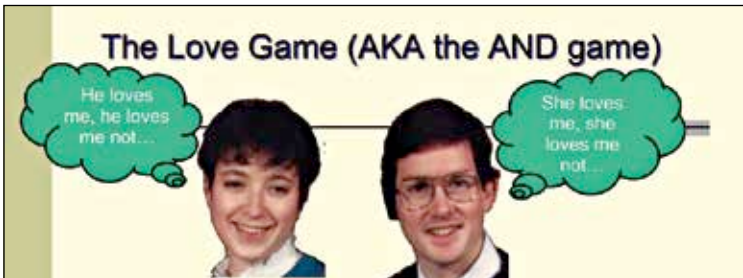


voting, sharing of signature or decryption functions, and situations that require private information retrieval. One popular application of secure MPC (multi-party computation) is solving the Yao's millionaire problem, i.e. two millionaires want to know which one of them is richer but without revealing their net worth to the other.

The millionaires' problem is a secure two-party computation problem. It has been generalized for multi-party computations. In a secure MPC, if no party can learn more from the public function and its result than what can it learn from its own input. For a better understanding of the concept, visit: <http://dgit.in/Qe37wP>



This concept is of great value in the field of cryptography. It has been proved that the multi-party computation problem can be solved if there exist unconditionally secure authenticated channels between pairs of participants. Consider four individuals Alice, Bob, Carol and Dave who want to calculate their average salary without revealing their salary to others. One way to calculate the salary is as follows:

- 1 Alice adds a secret random number to her salary, encrypts the result with Bob's public key and sends it to Bob.
- 2 Bob decrypts Alice's result with his private key. He adds his salary to Alice's message, encrypts the result with Carol's public key and sends it to Carol.
- 3 Carol decrypts Bob's result with her private key. She adds her salary to Bob's message, encrypts the result with Dave's public key and sends it to Dave.
- 4 Dave decrypts Carol's result with his private key. He adds his salary to Carol's message, encrypts the result with Alice's public key and sends it to Alice.